Piece rate ~

Utilization ~ $\frac{\text{time used}}{\text{time available}}$ $\boxed{= 99.2\%}$

42 workers 8 hours a day

16 minutes per garment

2,000 garments a day

$\frac{42}{55}$ $\boxed{76.4\%}$

55.8

$\frac{\text{mach ut}}{\text{Labor ut}}$ $\frac{76}{99.2} = <1$

Airline = $\frac{\text{Big}}{\text{small}}$

~~mason equation~~

~~mason ratio~~

$\boxed{2000}$ a day

15 days

throughput output rate

TPT x OR = WIP

Little's Law

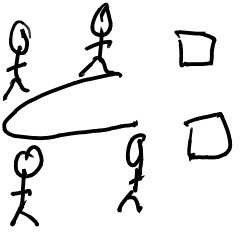
$\boxed{\text{Market}}$

Fixed

closer to customer demand =

go down

make this go down



1 2 3 4 5 6 7 8 9 a, b, c 10

1.2m

114 seconds

$\boxed{310}$ 31,208,71 seconds

344

114 x 12

Cycle Time

$\boxed{\text{Max places, but, agent time station}}$

5 7 5 2 3

Throughput time =

$\boxed{\text{Cycle time x stations}}$

.02% • .05 = .1% worker error

$238 = \frac{7.5 \times 60 \times 60}{114 \text{ sec}}$

Heterogen

→ Deterministic

vs

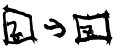
Variable → stochastic

output rate = cycle time

variable 3.5 ± 2.5

expected value

ave



50% < 2

Player

love ← body correction

food

work

discipline

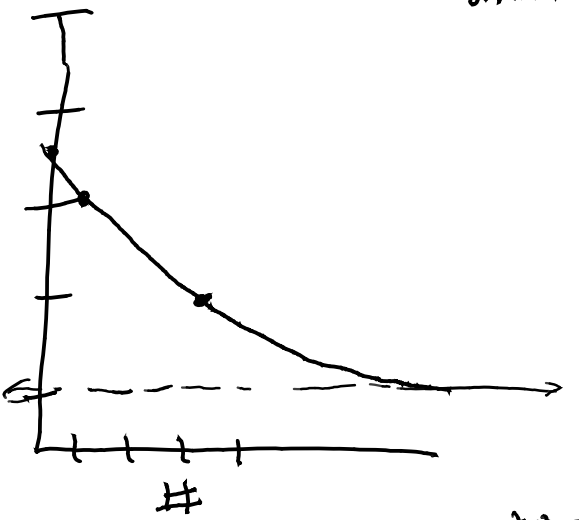
Blockage condition

1 goes faster than 2

Starvation condition

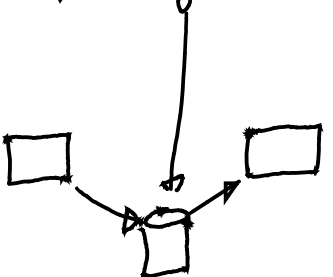
1 2 is faster than 2

Lean → force buffer to be small



inventory buffer — decouples two processes

creates independent process



Capacity — Productive

volume of WIP

$CAP = \frac{\text{units}}{\text{time}}$ inverse of cycle time $\frac{\text{time}}{\text{unit}}$

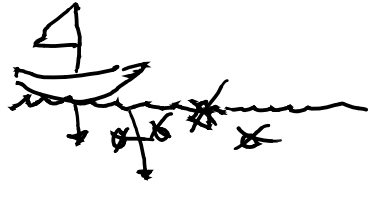
Weigh + test → K. Dump → Dump 1-16, Dump 17-24, Dump 25-27

18k.365

Restone

1 x 1500

1k.A / hour hours



Dr Money class

UG/IC class

ETA

Advanced modeling golf

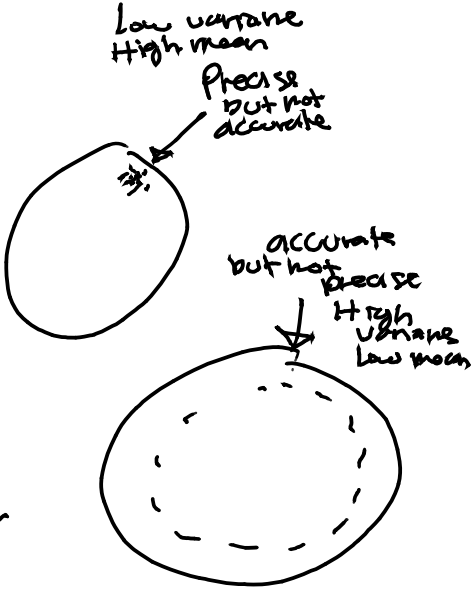
Upper

lower

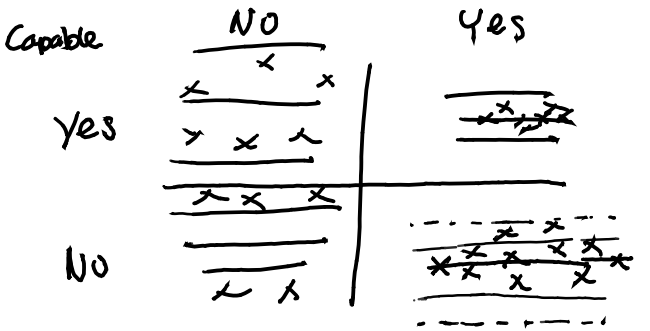
limits

come

from our



In control



6 Sigma

1. Voice of the customer

Oct 13th

sheet front and back of notes

what do I do

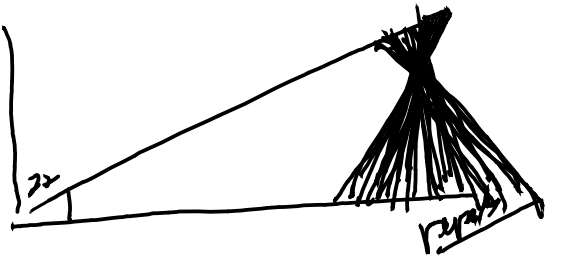
Meta Protein shake - 200

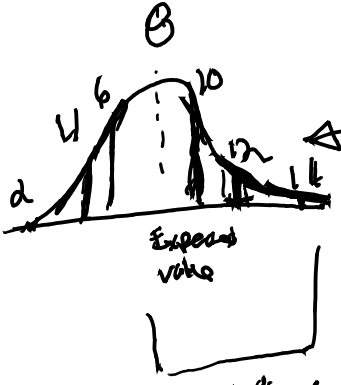
Delivered

1d

-1d

inventory carrying cost





Cost of understocking

* measure service level

our own cost of overstock or understock

90% of volume

3x

2x

1x

1x

1x

1x

1x

10

11

12

13

▽ [Cut] ▽ [Kit] ▽ [Stitch] ▽ [Steam] ▽ [LAST] ▽

TPT

WIP CT → 41 min per batch

CT

CT - cycle time - Karchunk time - Slowest step CT of one process

Throughput time - time it takes from start to finish

TPT

100 pairs/batch

Size .05 min

▽ → [Cut I] $(5.25+(.05\cdot100))4$ — 41 minutes

→ [Cut 2] $4(5+(.05\cdot100))$ — 40 min

→ [Cut 3] $4(4+(.04\cdot100))$ — 32 min

$\frac{480}{41}$ = Batches per day

Converted CT into output rate

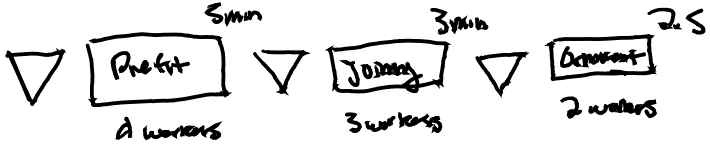
$\frac{\text{Time working}}{\text{Cycle time}}$ = output rate

Parallel process

↳ Process done at same time

$4(5.25+(.05\cdot10))$ → 20.1 batches

$=23$ 201 pair per day



$\frac{5\text{min}}{4 \text{ workers}}$

$\boxed{1.25 \text{ min per pair}}$ = Bottleneck

12 pairs per hours

48 pairs per hour

$\frac{3}{3}$

1 min per pair

$\frac{2.5}{2}$

$\boxed{1.25 \text{ min per pair}}$ = Bottleneck

384 pairs per day

41 min

10min

48 batches

125 min

continuous

6 hours

Start simple

add complexity

$CT \times WIP = TPT$

$\frac{1}{OR} \times WIP = TPT$

work in process inventory

$\frac{WIP}{OR} = TPT$

Output rate

40 sec

22

lambda — arrival rate

worker utilization

$$\rho = \frac{\lambda}{\mu}$$

M/M/1 queue

mew — service rate

Transformation

Book in bank

$$L_q = \frac{\rho^2}{(1-\rho)}$$

Myopic Optimization

Funnel for vasa

minimize our cost

JTD

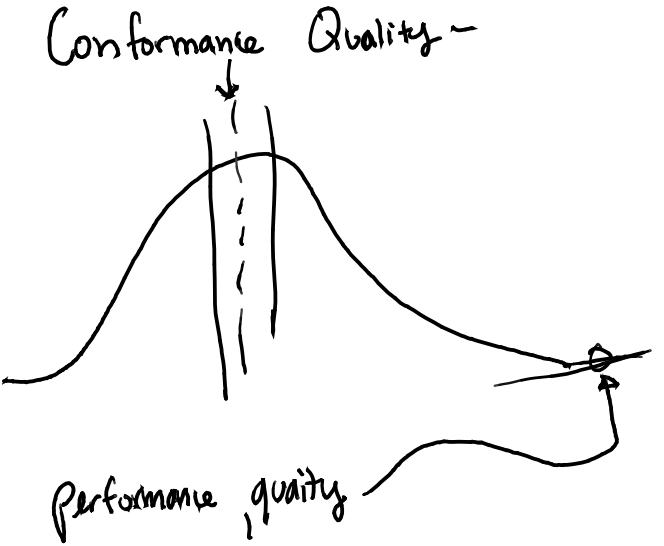
↳ Joint optimization

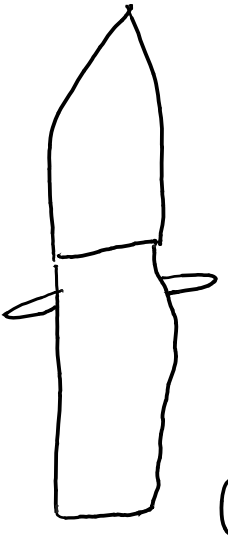
DAMPER

communicate

POS visibility

vertically integrate


Conformance Quality
performance quality



Grid fins
help steer

① Reas less dumb

② Delete the process steps
↳ tight margins

Simplify
optimize ← ③ Cut hedge your bets

* requirements need a person

⑥ Accelerate cycle time

⑧ Automate

Sell to seeing
through eyes of Love group

1. Take a screenshot

Personal Assistant

1.) Idea catchall, catch email & passwords! future todo items, things to learn

2.) Update calendar — via screenshots, emails etc

3.) Help me ace my classes
   a.) Remember & do my assignments
      i.) todoist matches 1:1 with canvas always
         1.) correct format & time
      ii.) Ensure assignments are planned and in the calendar
      iii.) follow up and make sure assignments get done.
   b.) Make sure I learn and master all material
      i.) understand the material enough to be able to teach each concept
      ii.) apply each concept to others
      iii.) Master material enough to get 100%.

4.) Maximize my time to achieve my other goals. Getting into Cougar Capital, Landing high paying job, Build a linkedin following, Reddit, Give-back finance app

EV = Equity + Debt

A)

EV = Multiples (EBITDA)

WF E = 14481 = x (1222)

WF 11.85 = x

Sprouts = 4102 + 304 = x (187)

S = 23.56 = x

Fresh Market = 1554 + 25 = x (138)

FM = 11.44 = x

$$WACC = \frac{E}{D+E} r_e + \frac{D}{D+E} r_d \cdot (1-T)$$

MRP = 5%

Capital Asset Pricing Model (CAPM) $\beta = .7$

30yr T bill = 3.4%

$$r_e = r_f + \beta(MRP)$$

$$r_e = r_f + \beta(r_m - r_f)$$

$$r = 3.4\% + .75(5\%)$$

$$\downarrow$$

$$3.5$$

$$r_e = WACC$$

3.

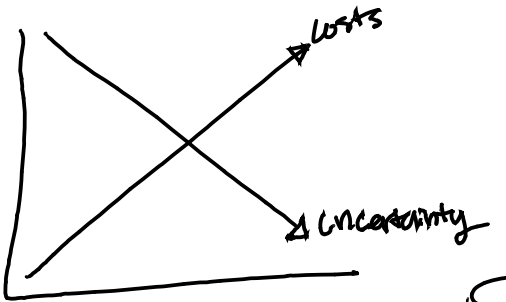
$\boxed{a.}$ $\boxed{6.a = WACC}$

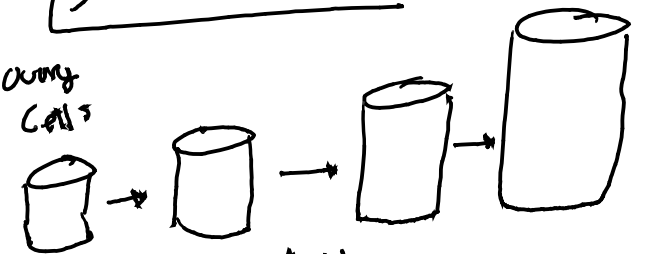
2.5%

CF = EBITDA

Vasa

- Re





12 kL
↓
4 kL media
↓
5.66 kg
↓
80% oocyte
↓
4.68 kg

Batches a year
15 x

↓

4.68 Raquen x 12 vessels

↓

42,400 g

↓

$\frac{42L}{.37g \text{ dose}}$

↓

2276,700 x 2000 doses a year

I pizza feeds kid in Africa

focus on healthy ing

catering with their own Brick oven

get food in 7 minutes

service operation

↳ I was there

Front and back double spaced.
you your family spirit job search

1. Bottlenecks
2. Balanced Processes
3. Hiccup